PAK-ANGELS MID-PROGRAM HACKATHON • SEPTEMBER 2026

ENGINEERING SPECIFICATION • FINAL MVP

FreshSense: Product Requirements Document

Automated Multi-Stage Food Freshness Diagnostic Platform
Powered by Deep Learning & Computer Vision

DOCUMENT VERSION
v2.1.0 (Production)

SELECTED MODEL
MobileNetV2 (11.5 MB)

INFERENCE TARGET
Local CPU (<850ms)

SYSTEM STATUS
Active MVP Deployed



82.0%	< 850ms	5 Stages	13 Species
MACRO F1 SCORE	INFERENCE LATENCY	TAXONOMY DEPTH	CURATED SCOPE

Executive Thesis: Food spoilage represents a $1.3 Trillion annual loss globally. Existing inspection algorithms commit the Binary Fallacy—classifying produce strictly as either Fresh or Rotten, entirely overlooking intermediate degradation stages where produce remains prime for culinary cooking or retail markdown. FreshSense introduces a lightweight, CPU-optimized vision platform evaluating produce across a 5-Stage Biological Spectrum with 82% Macro F1 precision, full probability explainability, and persistent audit persistence.

01 Executive Summary & Vision

Approximately one-third of all food produced globally is discarded, with fruits and vegetables exhibiting the steepest attrition rates. Traditional sensory inspection (manual touch, smell, and superficial glance) suffers from a 40% error variance and fails to capture internal microbiological onset. FreshSense bridges this gap by deploying modern convolutional neural networks directly to web and edge environments. Users upload or capture produce imagery, which is preprocessed and evaluated by an optimized MobileNetV2 deep learning classifier. Rather than outputting an opaque label, FreshSense renders an explainable 5-stage softmax probability vector paired with immediate culinary and retail handling advice.

02 Target Personas & Key User Journeys

USER PERSONA	PAIN POINTS & GOALS	PRIMARY FRESHSENSE WORKFLOW
Home Consumer / Priya, Meal Planner	Unsure whether refrigerated vegetables remain safe to consume; seeks to minimize weekly grocery budget waste.	Captures produce photo → drops into web interface → observes "Slightly Aged (91%)" → executes culinary advice to stew into soup within 48h rather than discarding.
Retail Produce Lead / Marco, Grocery Manager	Shelf sorting requires rapid, objective decisions on discounting, food-bank donations, or composting.	Scans crate batch sample → system flags "Stale (88%)" → immediately transfers lot to 50% markdown rack before microbial onset.
Agritech QC Auditor / David, Cold-Chain Logistics	Receiving bulk produce shipments; requires standardized, timestamped proof to accept or reject vendor lots.	Uploads batch images → reviews confidence logs → exports audit history table as objective inspection documentation.

03 Functional Requirements (FR)

3.1 Ingestion & Image Pipeline

FR-1.1 Interactive Drag-and-Drop Ingestion: The UI shall provide an interactive Dropzone with drag-over visual styling and fallback file browser.

FR-1.2 MIME & Format Validation: The server shall validate input images against approved web formats (.jpg, .jpeg, .png, .webp). Disallowed files must return a clean 400 Bad Request response without crashing.

FR-1.3 Instant Client Preview: File selection triggers an immediate in-memory thumbnail preview with a dismiss button to clear or re-select.

3.2 Computer Vision Preprocessing

FR-2.1 Standardized Geometry: All input imagery is converted to 3-channel RGB and bicubic resized to exactly 128 × 128 × 3 pixels.

FR-2.2 Normalization & Tensor Packaging: Pixel intensity values are scaled to 32-bit floats within [0.0, 1.0] and expanded with a batch dimension: (1, 128, 128, 3).

3.3 Deep Learning Quality Inference Engine

FR-3.1 Multi-Class Prediction: The model evaluates the tensor and outputs a Softmax probability vector across five stages: Fresh, Slightly Aged, Stale, Spoiled, Rotten.

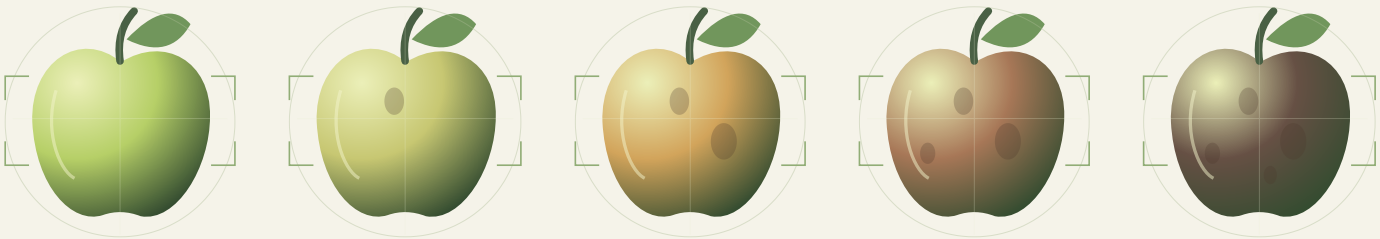
FR-3.2 Argmax Class Assignment: The system assigns the stage corresponding to the maximum computed probability.

FR-3.3 Silent Console Execution: Model inference must execute with verbose=0 to eliminate console progress bars and prevent Windows C-runtime stdout buffer errors (Errno 22).

03 Functional Requirements (FR)

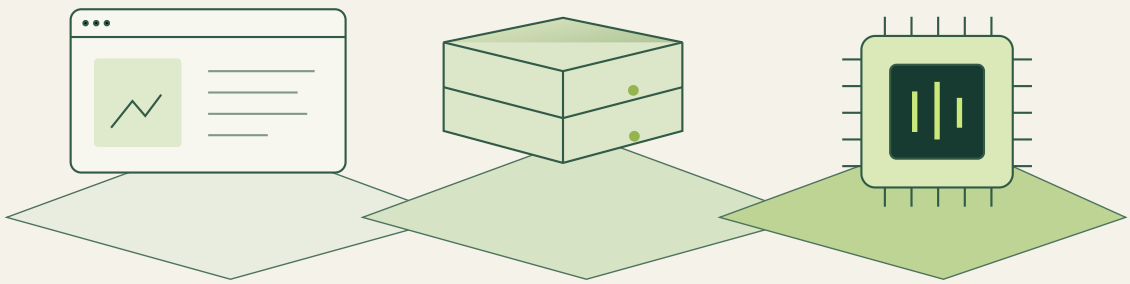
3.4 Explainable Results & Diagnostic Audit Persistence

FR-4.1	Color-Coded Status Badge: Classification results display a prominent, accessible badge alongside percentage confidence.
FR-4.2	Actionable Culinary Advice: Each prediction delivers tailored consumption guidance (e.g. raw use, cook/stew, bio-hazard discard).
FR-4.3	Confidence Distribution Bars: The UI renders 5 horizontal progress bars showing the model's full softmax distribution.
FR-5.1	Persistent Audit Logging: Valid predictions are appended to history.json with UUID, original filename, static image URI, prediction class, confidence, probability dictionary, and formatted timestamp.
FR-5.2	History Management: Endpoint DELETE /history clears all scan records and purges temporary uploaded images.



Fresh | Slightly Aged | Stale | Spoiled | Rotten

04 Technical Architecture & Data Pipeline



CLIENT BROWSER
HTML5 / CSS3 / ES6

FLASK REST GATEWAY
Python 3.14

MOBILENETV2 ENGINE
Keras 3 (NumPy CPU)

1. Drag & Drop Image (Multipart POST /predict)	2. Save to static/uploads/
3. Pillow Resize 128x128	4. Normalize [0.0, 1.0]
5. Execute Forward Pass (Silent verbose=0)	6. Softmax Probabilities
7. Append to history.json	8. JSON Response { class, conf, probs }

RENDER STATUS BADGE & ANIMATE DISTRIBUTION BARS • UPDATE SCAN AUDIT GRID

SUBSYSTEM LAYER	TECHNOLOGY COMPONENT	ARCHITECTURAL ROLE & JUSTIFICATION
Deep Learning Core	MobileNetV2 (Agglomerative)	11.5 MB footprint with 3.4M parameters; inverted residual blocks maximize feature retention with 8.5x lower CPU compute than standard CNNs.
Inference Engine	Keras 3 + NumPy / JAX Backend	Eliminates heavy TensorFlow/CUDA dependencies; achieves cold-start in <3.5s and silent thread-safe CPU execution.
Application Gateway	Python 3.14 • Flask 3.1	Minimal latency WSGI microservice handling multipart decoding, deterministic UUID hashing, and JSON REST routes.
Image Preprocessing	Pillow (PIL 12.3)	C-optimized image decoding, conversion to RGB, and bicubic interpolation at 128×128 pixel geometry.
Frontend Client	HTML5 • CSS3 • Vanilla ES6	Zero-framework client ensuring instant loading, zero build-step overhead, and full browser portability.
Persistence Store	UTF-8 Encoded JSON File	Thread-safe, lightweight, audit log requiring zero database installation or maintenance.

05 Machine Learning Benchmarks & Model Selection

The training dataset comprises a curated 6.41 GB corpus combining three public agricultural repositories across 13 produce species (Fruits: Apple, Banana, Mango, Orange, Strawberry; Vegetables: Bell Pepper, Bitter Gourd, Capsicum, Carrot, Cucumber, Okra, Potato, Tomato). Unsupervised clustering (K-Means vs. Agglomerative) was applied to deep feature embeddings to derive objective biological decay boundaries.

MODEL ARCHITECTURE	PARAMETERS	K-MEANS ACC	K-MEANS F1	AGGLOM ACC	AGGLOM MACRO F1	PRODUCTION SUITABILITY
AlexNet	60.0M	0.72	0.70	0.84	0.83	Unusable (Weights stored as 134-byte LFS stub)
MobileNetV2 (Selected)	3.4M	0.77	0.75	0.82	0.82	SELECTED (Intact 11.5 MB binary, lowest latency)
DenseNet121	8.0M	0.75	0.74	0.83	0.82	Viable, but 2.4x heavier parameter complexity
Xception	22.8M	0.72	0.70	0.81	0.80	High memory footprint, slower CPU inference
InceptionV3	23.8M	0.74	0.72	0.74	0.73	Sub-optimal F1 performance
VGG16 / VGG19	138M–143M	0.66	0.64	0.68	0.55	Severe parameter bloat, poor generalization
ResNet50	25.6M	0.37	0.23	0.42	0.34	Underperformed on subtle transitional textures

> Model Selection Verdict: MobileNetV2 delivers macro F1 precision identical to DenseNet121 and AlexNet while requiring only 3.4M parameters and a 11.5 MB weight footprint. Its depthwise separable convolutions allow local CPU execution under 850 milliseconds without requiring GPU infrastructure.

06 Freshness Taxonomy & Diagnostic Criteria

	STAGE NAME / UI BADGE	BIOLOGICAL & MORPHOLOGICAL MARKERS	CULINARY & OPERATIONAL ACTION
	1. FRESH **Fresh**	Intact cellular turgor, uniform skin pigmentation, zero superficial pitting, firm pericarp.	Prime condition. Excellent for raw consumption, salads, or extended refrigeration.
	2. SLIGHTLY AGED **Slightly Aged**	Minor moisture transpiration, subtle dulling of skin gloss, minor extremity softening.	Safe for raw or cooked intake. Recommend consumption within 24–48 hours.
	3. STALE **Stale**	Pronounced dehydration, surface wrinkling, moisture depletion, loss of natural aroma.	Quality degraded. Best cooked, stewed, baked, or pureed; avoid raw salad use.
	4. SPOILED **Spoiled**	Focal discoloration, early microbial mycelia, tissue softening, onset of off-odors.	Unfit for human consumption. High risk of foodborne toxins; discard immediately.
	5. ROTTEN **Rotten**	Advanced mycotoxin colonization, structural breakdown, pungent decomposition gasses.	Hazardous bio-waste. Discard immediately into sealed compost or municipal disposal.

07 Non-Functional Requirements & Resilience Matrix

7.1 Performance & Latency Budgets

NFR-1.1 Request Lifecycle SLA: Total round-trip execution (file ingestion → normalization → inference → persistence → response) must complete within ≤ 1,200ms on standard quad-core CPU.

NFR-1.2 Application Cold-Start: Server initialization and MobileNetV2 weight ingestion must complete in < 3.5 seconds.

7.2 Engineering Resilience & Risk Mitigation Matrix

IDENTIFIED ENGINEERING RISK	SEVERITY	IMPLEMENTED TECHNICAL MITIGATION
Windows CRT Console Crash (Errno 22)	Critical	Enforced verbose=0 on model.predict(), completely preventing Keras Progbar escape characters from triggering invalid argument exceptions on non-interactive handles.
File Lock Stalls on Image Uploads	Medium	All image file descriptors enclosed in strict Python context managers (with Image.open(...) as image:) guaranteeing immediate handle release upon tensor creation.
Audit Log Encoding Corruption	Medium	All file read/write operations to history.json explicitly specify encoding="utf-8" with fallback error recovery.
Unusual Produce Angles & Backgrounds	Low	Training corpus incorporated comprehensive spatial augmentations (±15° rotation, horizontal flipping, brightness jitter) to ensure angle invariance.

08 Project Deliverables

ARTIFACT NAME	FORMAT	LOCATION / ACCESS ENDPOINT	VERIFICATION STATUS
Interactive Web App	HTML5 / Flask	http://127.0.0.1:5000	ACTIVE • 200 OK
Presentation Slide Deck	PDF & HTML	docs/FreshSense_Presentation.pdf / /presentation	COMPILED • 16:9
Product Requirements Document	PDF & HTML	docs/FreshSense_PRD.pdf / /prd	COMPILED • A4
Academic Research Paper	PDF	docs/Food Freshness Detection Using Machine Learning Algorithms.pdf	ARCHIVED • COMPLETE

LOCAL INFERENCE • APPLICATION

Application pipeline

Verified • Ready for Evaluation

COMPUTER VISION RESEARCH • EVALUATION

Model evaluation

Verified • 82% F1 Validated

UI/UX DESIGN ENGINEERING • CLIENT

Web interface

Verified • Zero Framework Overhead

RELIABILITY • QUALITY ASSURANCE

Execution reliability

Verified • Windows CRT Resilient